

JPMethod

Long Division

American bracket notation - 3-digit / 2-digit to 4-digit / 3-digit

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Long Division - American bracket notation - 3-digit / 2-digit to 4-digit / 3-digit Instructions

American long division: divisor on the left, dividend inside the bracket, quotient above the vinculum. Subtract, bring down the next digit, repeat.

- Start with Section 1 - all divide evenly. Build the muscle memory first.
- Section 2 introduces remainders. Write answers as 'q r N'.
- Section 3 steps up to a 4-digit dividend. Same algorithm, one more bring-down.
- Section 4 is the stretch - 4-digit dividends with 3-digit divisors. Estimate before you multiply.
- Use the multiples reference table on the inside cover if estimation is hard.
- Check your work: quotient * divisor + remainder should equal the dividend.

Long division rewards patience. Show every step in the bracket layout - it makes mistakes visible and easy to fix.

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How American long division works

1. Set up: divisor on the LEFT, dividend INSIDE the bracket, quotient ABOVE the bar.

2. Look at the dividend from the left. Take just enough digits so the value is at least as big as the divisor.

3. Divide. Estimate how many times the divisor fits. Write that digit ABOVE the bar, aligned with the rightmost digit you used.

4. Multiply your quotient digit by the divisor. Write the result **13 r 5** BELOW the digits you used.

5. Subtract. Draw a short line and write the difference underneath.

6. Bring down the next digit of the dividend. The result becomes the next number to divide.

7. Repeat: divide, multiply, subtract, bring down. Whatever is left at the end is the REMAINDER.

8. Check: quotient x divisor + remainder should equal the dividend.

Example: $200 \div 15$

$$\begin{array}{r}
 13 \\
 \hline
 15 \overline{) 200} \\
 \underline{15} \\
 50 \\
 \underline{45} \\
 5
 \end{array}$$

Multiples reference (divisor x 1..9)

Cuts estimation time. Memorise the rows for whichever divisor you find hardest.

x	1	2	3	4	5	6	7	8	9
12	12	24	36	48	60	72	84	96	108
13	13	26	39	52	65	78	91	104	117
14	14	28	42	56	70	84	98	112	126
15	15	30	45	60	75	90	105	120	135
16	16	32	48	64	80	96	112	128	144
17	17	34	51	68	85	102	119	136	153
18	18	36	54	72	90	108	126	144	162
19	19	38	57	76	95	114	133	152	171
20	20	40	60	80	100	120	140	160	180
25	25	50	75	100	125	150	175	200	225

Section 1. 3-digit / 2-digit, exact

Each of these divides evenly. No remainder.

1.

$$12 \overline{) 144}$$

2.

$$13 \overline{) 169}$$

3.

$$14 \overline{) 196}$$

4.

$$15 \overline{) 225}$$

5.

$$16 \overline{) 288}$$

6.

$$17 \overline{) 272}$$

Section 2. 3-digit / 2-digit, with remainder

Some have a remainder. Write the answer as 'q r N'.

1.

$$15 \overline{) 200}$$

2.

$$25 \overline{) 370}$$

3.

$$12 \overline{) 160}$$

4.

$$13 \overline{) 147}$$

5.

$$16 \overline{) 196}$$

6.

$$17 \overline{) 285}$$

Section 2. 3-digit / 2-digit, with remainder

Some have a remainder. Write the answer as 'q r N'.

7.

$$18 \overline{) 430}$$

8.

$$21 \overline{) 555}$$

9.

$$23 \overline{) 612}$$

10.

$$24 \overline{) 777}$$

11.

$$19 \overline{) 840}$$

12.

$$22 \overline{) 901}$$

Section 3. 4-digit / 2-digit

Mix of exact and remainder. Show every step.

1.

$$16 \overline{) 1024}$$

2.

$$25 \overline{) 1500}$$

3.

$$23 \overline{) 2024}$$

4.

$$14 \overline{) 3500}$$

5.

$$18 \overline{) 1296}$$

6.

$$49 \overline{) 2401}$$

Section 4. 4-digit / 3-digit (stretch)

Mix of exact and remainder. Show every step.

1.

$$125 \overline{) 1500}$$

2.

$$150 \overline{) 2400}$$

3.

$$168 \overline{) 3528}$$

4.

$$333 \overline{) 9999}$$

5.

$$256 \overline{) 4096}$$

6.

$$189 \overline{) 8505}$$

Section 1. 3-digit / 2-digit, exact

1.

$$\begin{array}{r} 12 \overline{) 144} \\ \underline{12} \\ 24 \\ \underline{24} \\ 0 \end{array}$$

= 12

2.

$$\begin{array}{r} 13 \overline{) 169} \\ \underline{13} \\ 39 \\ \underline{39} \\ 0 \end{array}$$

= 13

3.

$$\begin{array}{r} 14 \overline{) 196} \\ \underline{14} \\ 56 \\ \underline{56} \\ 0 \end{array}$$

= 14

4.

$$\begin{array}{r} 15 \overline{) 225} \\ \underline{15} \\ 75 \\ \underline{75} \\ 0 \end{array}$$

= 15

Section 1. 3-digit / 2-digit, exact

5.

$$\begin{array}{r}
 16 \overline{) 288} \\
 \underline{16} \\
 128 \\
 \underline{128} \\
 0
 \end{array}$$

= 18

6.

$$\begin{array}{r}
 17 \overline{) 272} \\
 \underline{17} \\
 102 \\
 \underline{102} \\
 0
 \end{array}$$

= 16

Section 2. 3-digit / 2-digit, with remainder

1.

$$\begin{array}{r} 13 \\ 15 \overline{) 200} \\ \underline{15} \\ 50 \\ \underline{45} \\ 5 \end{array}$$

= 13 r 5

2.

$$\begin{array}{r} 14 \\ 25 \overline{) 370} \\ \underline{25} \\ 120 \\ \underline{100} \\ 20 \end{array}$$

= 14 r 20

3.

$$\begin{array}{r} 13 \\ 12 \overline{) 160} \\ \underline{12} \\ 40 \\ \underline{36} \\ 4 \end{array}$$

= 13 r 4

4.

$$\begin{array}{r} 11 \\ 13 \overline{) 147} \\ \underline{13} \\ 17 \\ \underline{13} \\ 4 \end{array}$$

= 11 r 4

Section 2. 3-digit / 2-digit, with remainder

5.

$$\begin{array}{r}
 16 \overline{) 196} \\
 \underline{16} \\
 36 \\
 \underline{32} \\
 4
 \end{array}$$

= 12 r 4

6.

$$\begin{array}{r}
 17 \overline{) 285} \\
 \underline{17} \\
 115 \\
 \underline{102} \\
 13
 \end{array}$$

= 16 r 13

7.

$$\begin{array}{r}
 18 \overline{) 430} \\
 \underline{36} \\
 70 \\
 \underline{54} \\
 16
 \end{array}$$

= 23 r 16

8.

$$\begin{array}{r}
 21 \overline{) 555} \\
 \underline{42} \\
 135 \\
 \underline{126} \\
 9
 \end{array}$$

= 26 r 9

Section 2. 3-digit / 2-digit, with remainder

9.

$$\begin{array}{r} 26 \\ 23 \overline{) 612} \\ \underline{46} \\ 152 \\ \underline{138} \\ 14 \end{array}$$

$$= 26 \text{ r } 14$$

10.

$$\begin{array}{r} 32 \\ 24 \overline{) 777} \\ \underline{72} \\ 57 \\ \underline{48} \\ 9 \end{array}$$

$$= 32 \text{ r } 9$$

11.

$$\begin{array}{r} 44 \\ 19 \overline{) 840} \\ \underline{76} \\ 80 \\ \underline{76} \\ 4 \end{array}$$

$$= 44 \text{ r } 4$$

12.

$$\begin{array}{r} 40 \\ 22 \overline{) 901} \\ \underline{88} \\ 21 \\ \underline{0} \\ 21 \end{array}$$

$$= 40 \text{ r } 21$$

Section 3. 4-digit / 2-digit

1.

$$\begin{array}{r}
 \overline{64} \\
 16 \overline{) 1024} \\
 \underline{96} \\
 64 \\
 \underline{64} \\
 0
 \end{array}$$

= 64

2.

$$\begin{array}{r}
 \overline{60} \\
 25 \overline{) 1500} \\
 \underline{150} \\
 0 \\
 \underline{0} \\
 0
 \end{array}$$

= 60

3.

$$\begin{array}{r}
 \overline{88} \\
 23 \overline{) 2024} \\
 \underline{184} \\
 184 \\
 \underline{184} \\
 0
 \end{array}$$

= 88

4.

$$\begin{array}{r}
 \overline{250} \\
 14 \overline{) 3500} \\
 \underline{28} \\
 70 \\
 \underline{70} \\
 0 \\
 \underline{0} \\
 0
 \end{array}$$

= 250

Section 3. 4-digit / 2-digit

5.

$$\begin{array}{r}
 \overline{72} \\
 18 \overline{) 1296} \\
 \underline{126} \\
 36 \\
 \underline{36} \\
 0
 \end{array}$$

= 72

6.

$$\begin{array}{r}
 \overline{49} \\
 49 \overline{) 2401} \\
 \underline{196} \\
 441 \\
 \underline{441} \\
 0
 \end{array}$$

= 49

Section 4. 4-digit / 3-digit (stretch)

1.

$$\begin{array}{r}
 \overline{12} \\
 125 \overline{) 1500} \\
 \underline{125} \\
 250 \\
 \underline{250} \\
 0
 \end{array}$$

= 12

2.

$$\begin{array}{r}
 \overline{16} \\
 150 \overline{) 2400} \\
 \underline{150} \\
 900 \\
 \underline{900} \\
 0
 \end{array}$$

= 16

3.

$$\begin{array}{r}
 \overline{21} \\
 168 \overline{) 3528} \\
 \underline{336} \\
 168 \\
 \underline{168} \\
 0
 \end{array}$$

= 21

4.

$$\begin{array}{r}
 \overline{30} \\
 333 \overline{) 9999} \\
 \underline{999} \\
 9 \\
 \underline{0} \\
 9
 \end{array}$$

= 30 r 9

Section 4. 4-digit / 3-digit (stretch)

5.

$$\begin{array}{r}
 \overline{16} \\
 256 \overline{) 4096} \\
 \underline{256} \\
 1536 \\
 \underline{1536} \\
 0
 \end{array}$$

= 16

6.

$$\begin{array}{r}
 \overline{45} \\
 189 \overline{) 8505} \\
 \underline{756} \\
 945 \\
 \underline{945} \\
 0
 \end{array}$$

= 45